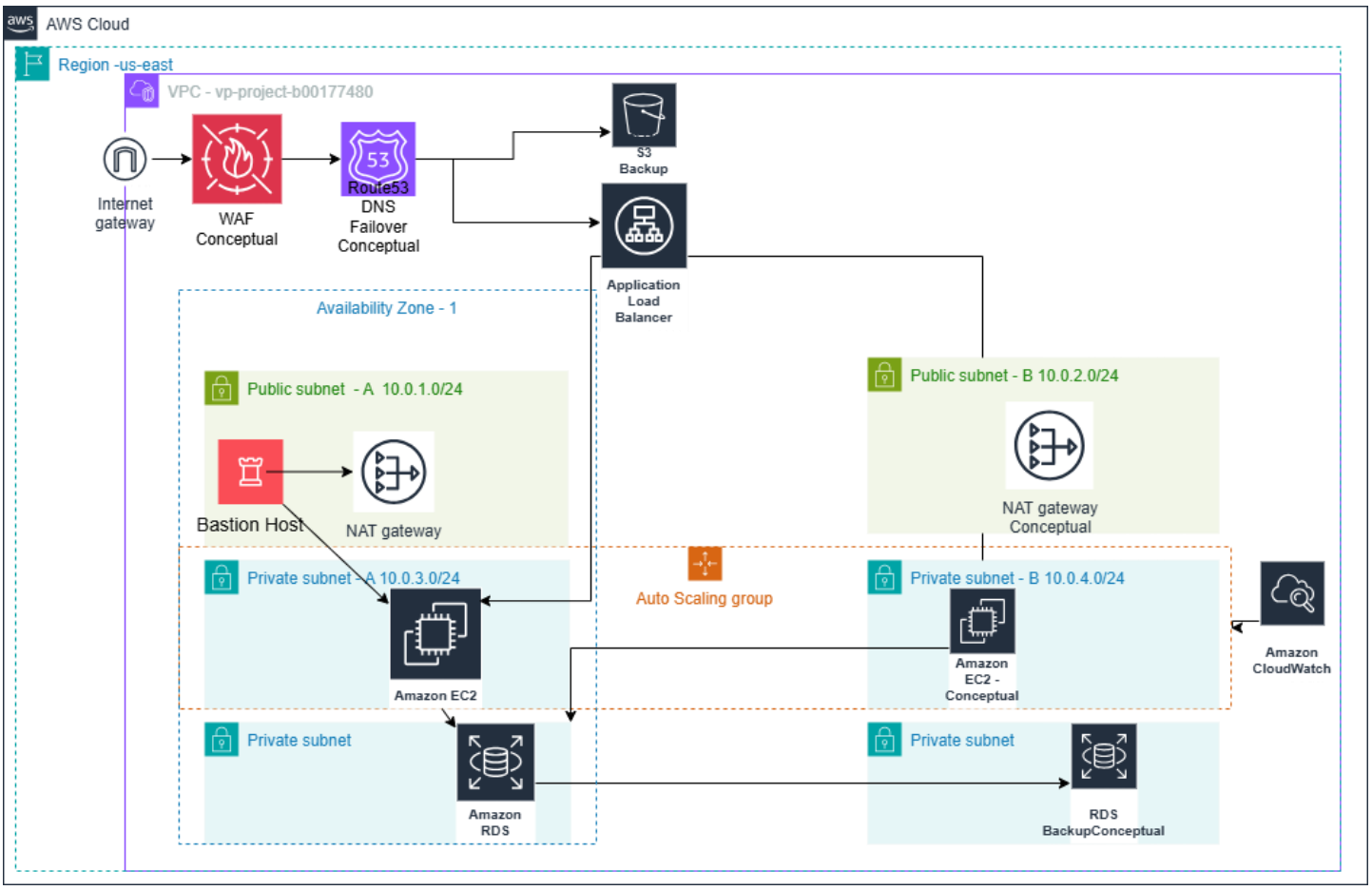


Following screenshot provides resource map of the VPC for WinLocal.

An Internet Gateway was attached to the VPC to allow inbound internet traffic to reach the public subnets.

A Regional NAT Gateway was deployed in the public subnet to allow private EC2 instances to reach the internet for software updates and patches without being directly exposed to inbound internet traffic.

An S3 Gateway VPC Endpoint was also configured, allowing EC2 instances to access S3 directly through the AWS network without routing through the NAT Gateway, reducing cost and improving security.

Two public and two private subnets were created, this number was chosen to ensure higher availability and fault tolerance. Additionally, and RDS subnet was created for the Database.

Two route tables were configured for both public subnets and two for both private subnets.



Name	Security group ID	Security group name	VPC ID	Description	Owner
–	sg-016357368355a9b05	bastion-host-vpc-b00177480	vpc-0b53be91388a377ea	bastion-host-vpc created 2026-05-01T...	499232271564
–	sg-05fa081c99cb39973	secgroup-webserver-b00177480	vpc-0b53be91388a377ea	secgroup-webserver 2026-05-01T19:24...	499232271564
–	sg-08f2ca9ac35401438	ec2-rds-1	vpc-0b53be91388a377ea	Security group attached to instances to ...	499232271564
–	sg-023240a9c65259a30	rds-ec2-1	vpc-0b53be91388a377ea	Security group attached to WinLocal-R...	499232271564
–	sg-03301398d2eaabd2f	RDS-security-group	vpc-0b53be91388a377ea	Allow SQL connection from private sub...	499232271564
–	sg-0d2370dedd7e98e67	secgroup-apploadbalancer-b00177480	vpc-0b53be91388a377ea	Allow HTTP and HTTPS to LoadBalancer	499232271564
–	sg-0d95b01e456c7e73b	default	vpc-0b53be91388a377ea	default VPC security group	499232271564

Security groups were created for bastion, RDS, webserver, Application load balancer.

Bastion host is created to allow configuration of webserver only via ssh on port 22 from Bastion host.

Only admin IP address is allowed to connect to it and connect to webserver from it.

```
   ~\_  ####_        Amazon Linux 2023
  ~~  \_#####\
  ~~     \###|
  ~~       \#/ ___   https://aws.amazon.com/linux/amazon-linux-2023
   ~~       V~' '->
    ~~~         /
      ~~._.   _/
         _/ _/
       _/m/'
Last login: Fri May  1 19:54:46 2026 from 89.101.46.77
[ec2-user@ip-10-0-1-160 ~]$ ls
bastionkey-b00177480.pem
[ec2-user@ip-10-0-1-160 ~]$ chmod 400 bastionkey-b00177480.pem"
>
> q
> exit
>
> ^C
[ec2-user@ip-10-0-1-160 ~]$ chmod 400 bastionkey-b00177480.pem"

q
exit
^C
[ec2-user@ip-10-0-1-160 ~]$ chmod 400 bastionkey-b00177480.pem
[ec2-user@ip-10-0-1-160 ~]$ ssh -i "bastionkey-b00177480.pem" ec2-user@10.0.3.29
       #_
   ~\_  ####_        Amazon Linux 2023
  ~~  \_#####\
  ~~     \###|
  ~~       \#/ ___   https://aws.amazon.com/linux/amazon-linux-2023
   ~~       V~' '->
    ~~~         /
      ~~._.   _/
         _/ _/
       _/m/'
[ec2-user@ip-10-0-3-29 ~]$
```

```
[ec2-user@ip-10-0-3-29 ~]$ mysql -h winlocal-rds-b00177480.ck0xaxobxtt4.us-east-1.rds.amazonaws.com -P 3306 -u admin -p
Enter password:
Welcome to the MariaDB monitor.  Commands end with ; or \g.
Your MySQL connection id is 28
Server version: 8.4.8 Source distribution

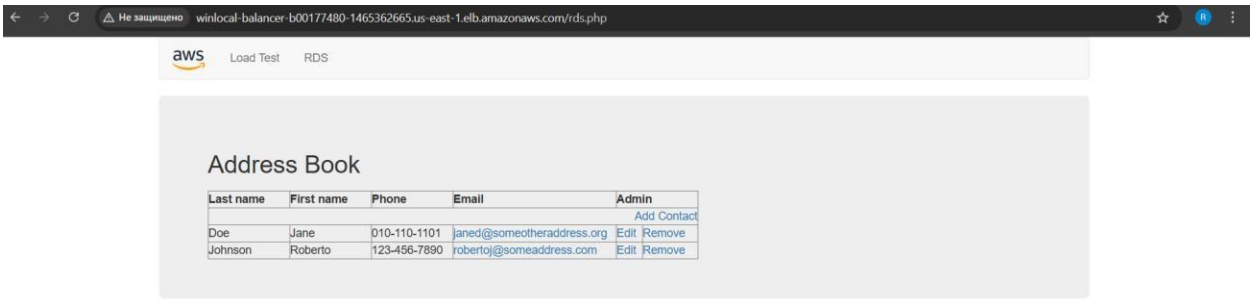
Copyright (c) 2000, 2018, Oracle, MariaDB Corporation Ab and others.

Type 'help;' or '\h' for help. Type '\c' to clear the current input statement.

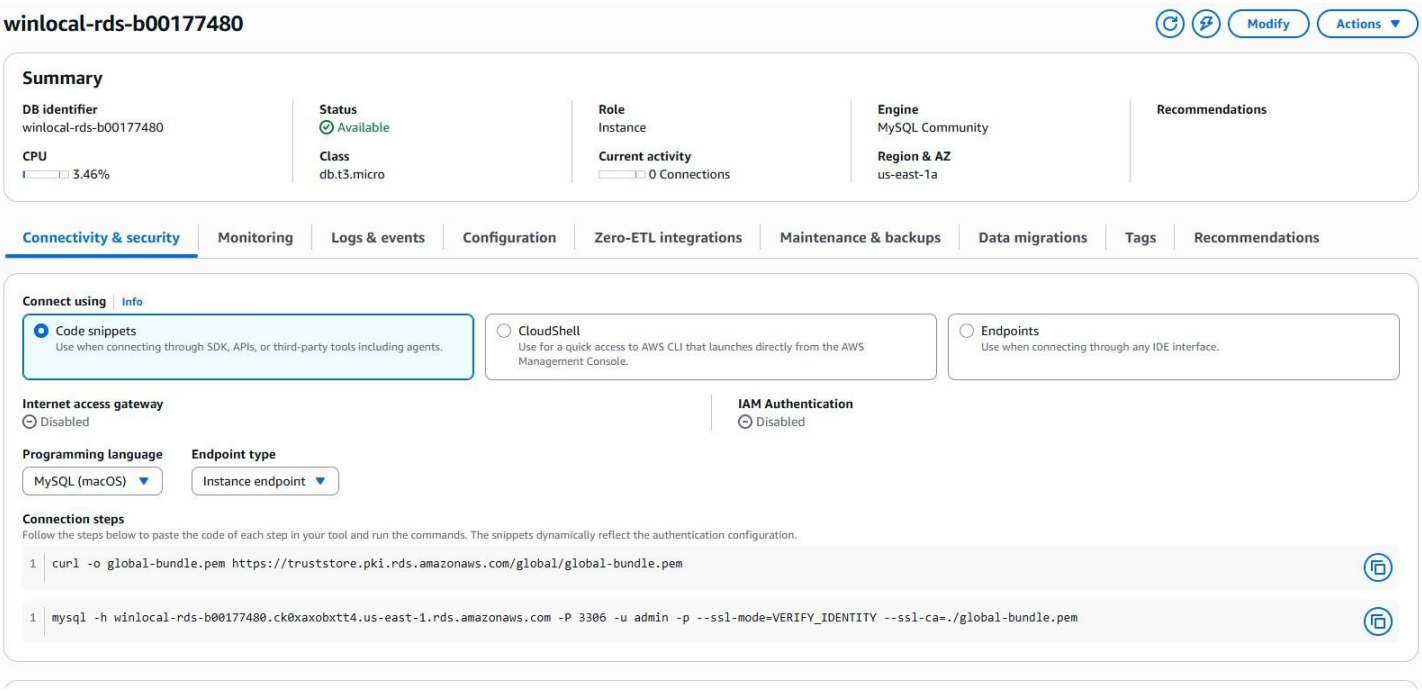
MySQL [(none)]> show databases;
+--------------------+
| Database           |
+--------------------+
| information_schema |
| mysql              |
| performance_schema |
| sys                |
| winlocal           |
+--------------------+
5 rows in set (0.007 sec)

MySQL [(none)]>
```

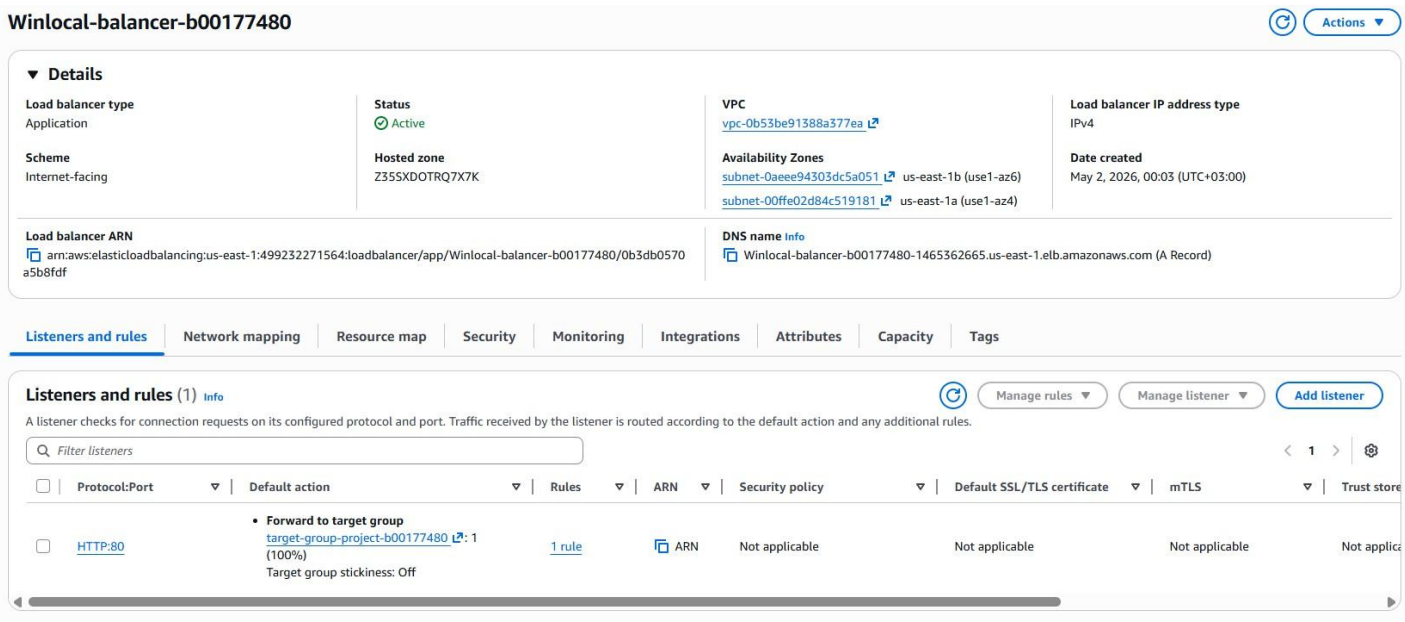
Screenshots are showing working MySQL database on the webserver and connection to it from Bastion.

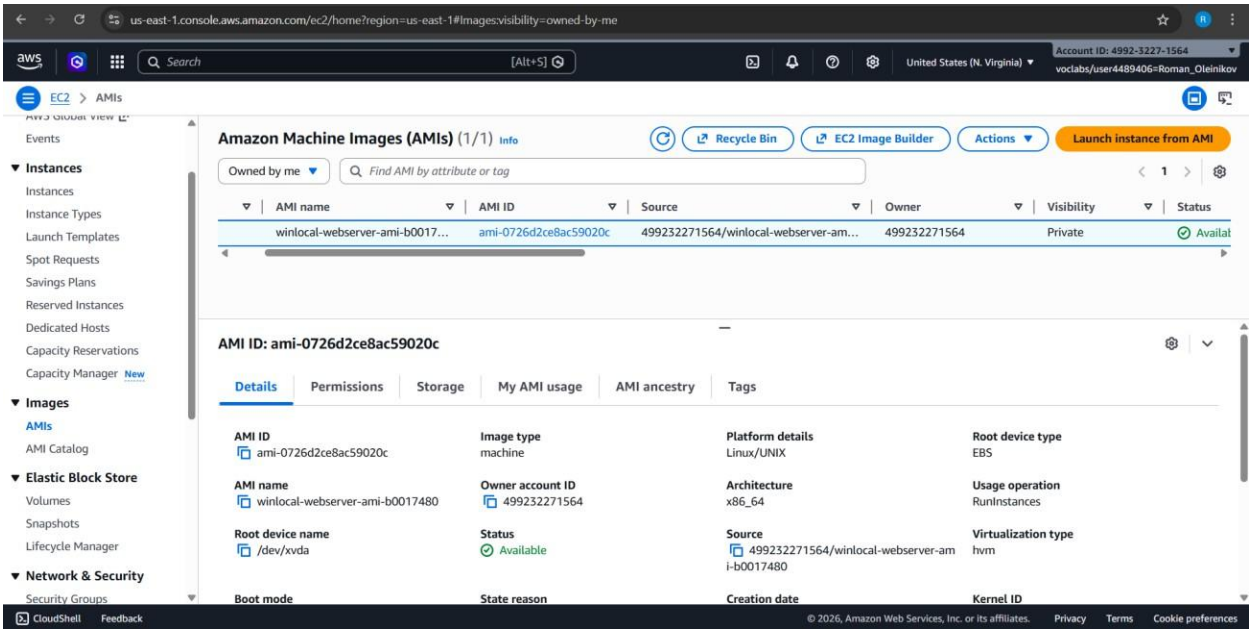


RDS was created to store competition entries including user names and addresses. Amazon RDS MySQL was selected as it is a fully managed relational database service that handles routine database tasks such as provisioning, patching, backup and recovery automatically



Here we can see working website, which is available from the Application load balancer, it is allowed to receive HTTP and HTTPS connection and connect to WebServer instance via those protocols.

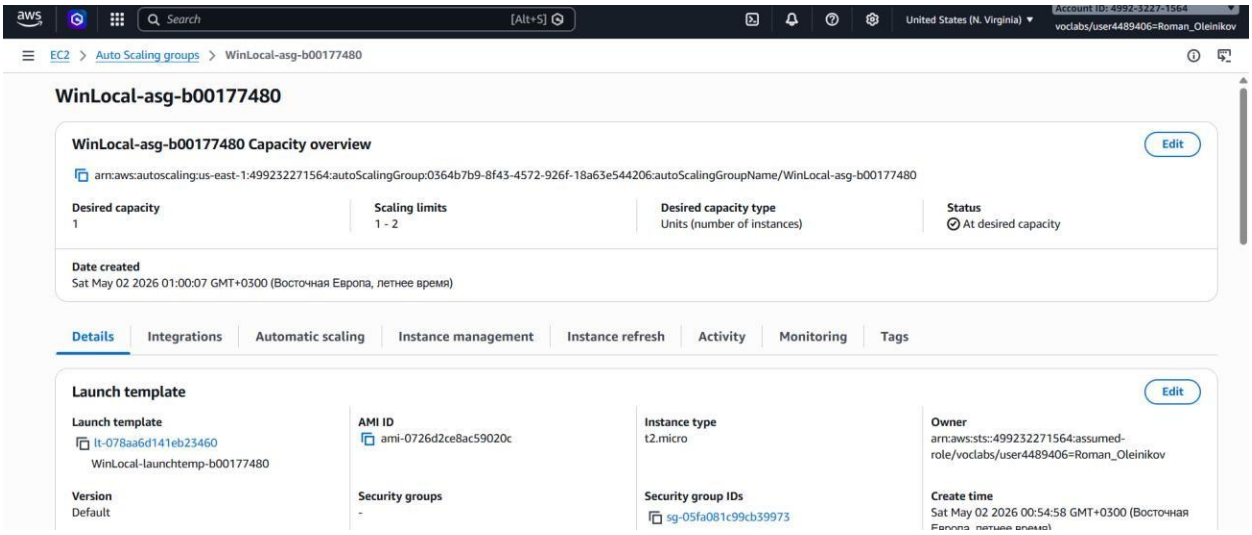


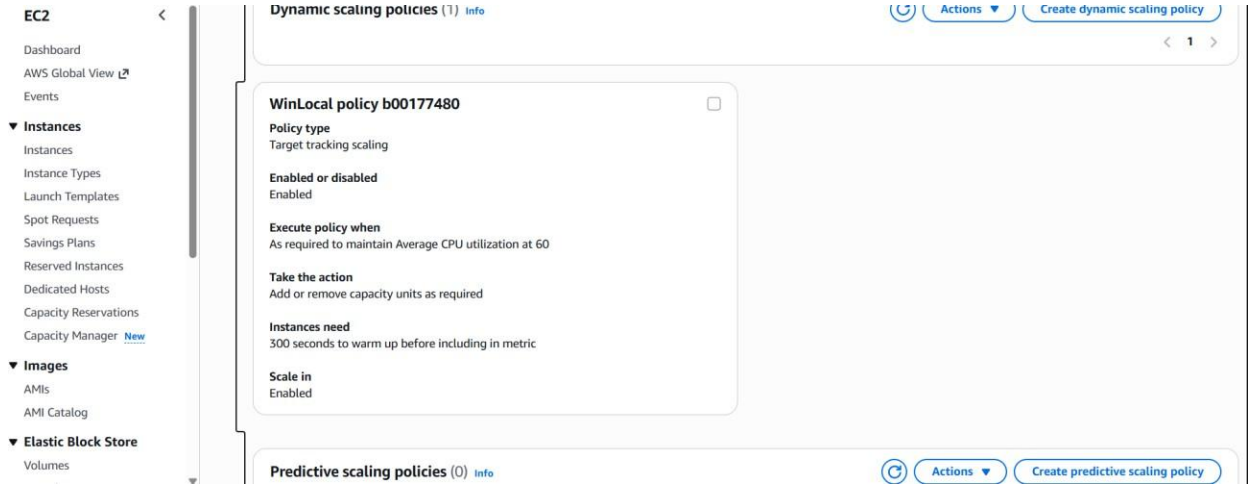


AMI was created for the webserver instance. An Amazon Machine Image is a pre-configured snapshot of a running EC2 instance that captures the operating system, installed software, application files and configuration at a specific point in time. Once created, the AMI can be used to launch new identical instances instantly without any manual configuration.

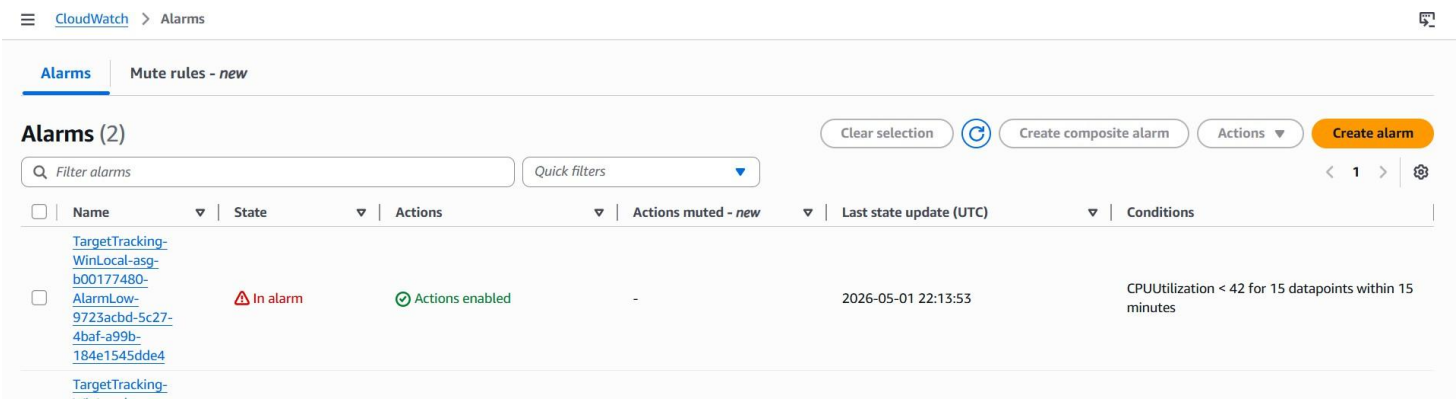
For WinLocal, AMIs provide two critical benefits. First, they enable the Auto Scaling Group to automatically launch pre-configured web server instances during traffic spikes — each new instance launched from the AMI comes up immediately with Apache, PHP and the WinLocal application already installed and configured, ready to serve traffic without any manual intervention. Second, AMIs support disaster recovery — if a running instance fails, a replacement can be launched from the AMI within minutes, significantly reducing recovery time.

Auto Scaling Group was implemented with policies as required by the company.





Tests were conducted to ensure that the CloudWatch would trigger alarm if conditions are met.



Conceptually, there are two more services to implement.

WinLocal recently suffered a ransomware attack where malicious actors gained access to the web application and compromised the backend database. WAF would be a good addition to existing security measures to filter out all suspicious and unvanted traffic before it reaches the VPC.

Route 53 failover is necessary to allow for automatic redirect to S3 bucket in case health check are not passed. It ensures that even in a worst-case scenario users receive a clear branded message rather than a broken experience, protecting WinLocal's reputation.